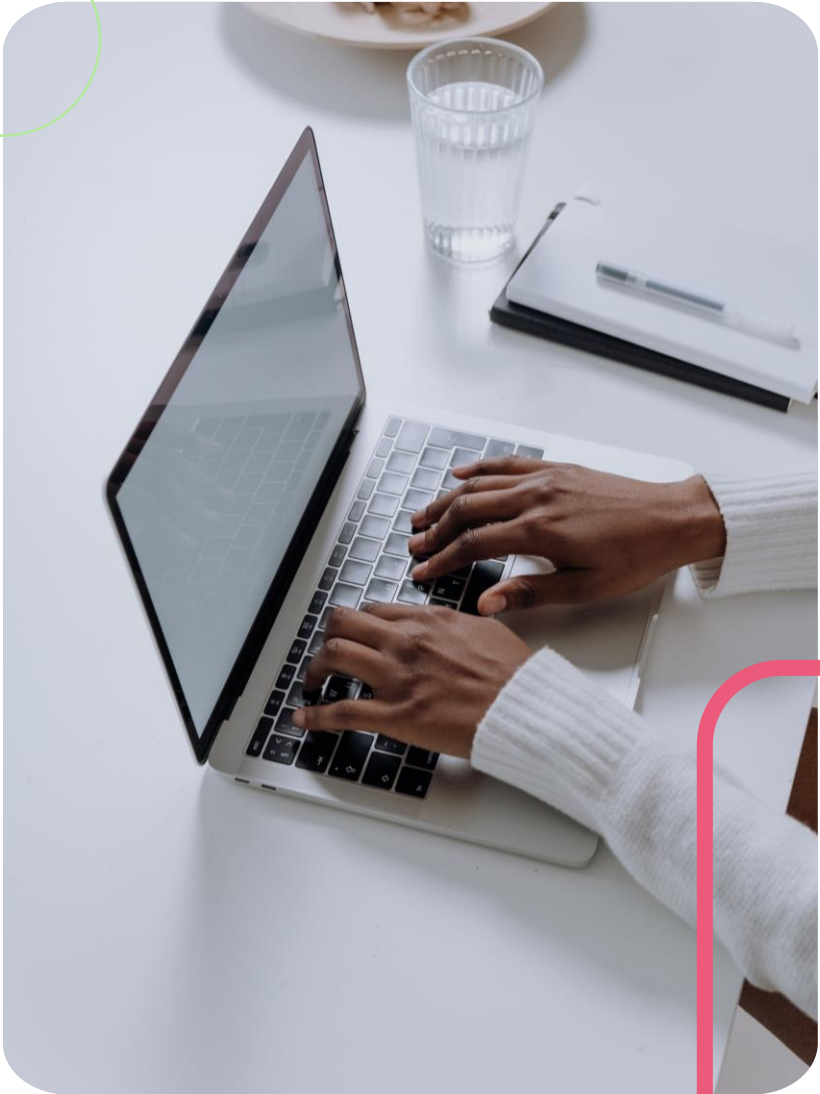




Diploma in

Data Analytics

Lesson 8: Logistic regression



Logistic regression <

Fit statistics <

Recap Module 2 <

Objectives

Lesson 7



Logistic regression

Recap logistic regression

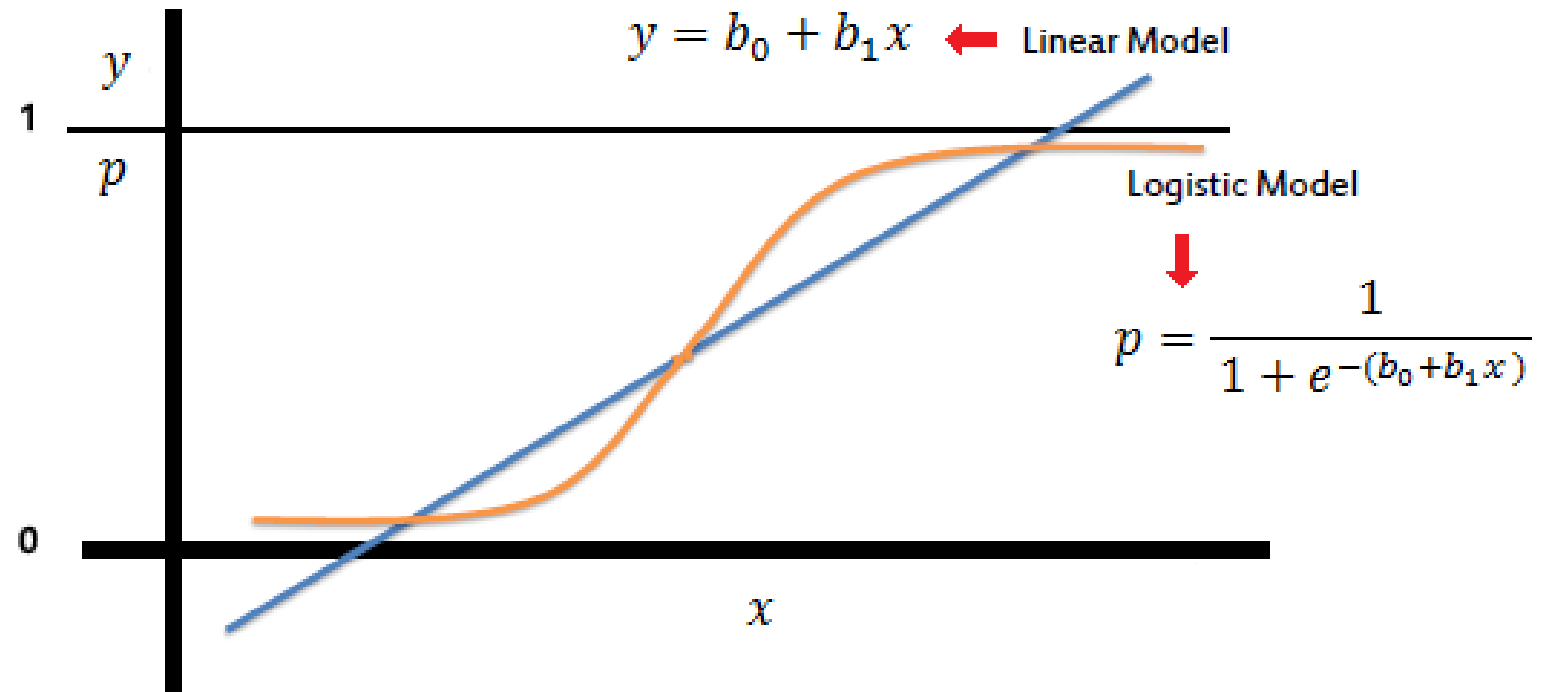
- Categorical response variable
 - 2 levels
 - 0, 1
- S-shaped curve



S-shaped curve

Probability between 0 and 1

Unlike linear regression, predictor variables does not have to be normally distributed or have equal variances between groups



Source: https://saedsayad.com/logistic_regression.htm

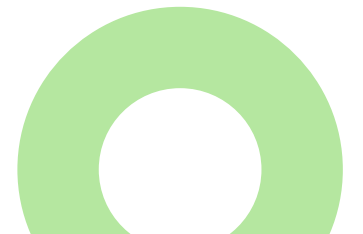


Logistic regression mathematically

$$p = \frac{1}{1 + e^{-(b_0 + b_1 x)}}$$

$$\frac{p}{1 - p} = \exp(b_0 + b_1 x)$$

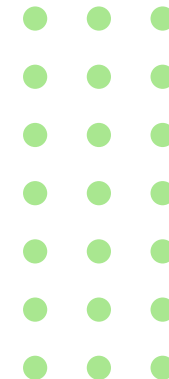
$$\ln\left(\frac{p}{1 - p}\right) = b_0 + b_1 x$$



Logistic regression examples

- Banking industry
 - Will client default on loan?
 - Is this a good client to take on?
- Medicine
 - Tumour benign or malignant





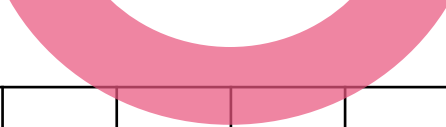
Did
you
know?

Linear vs logistic

Ordinary least square regression is used to estimate the line of best fit for linear regression.

Maximum likelihood estimation is used to estimate coefficient for logistic regression.





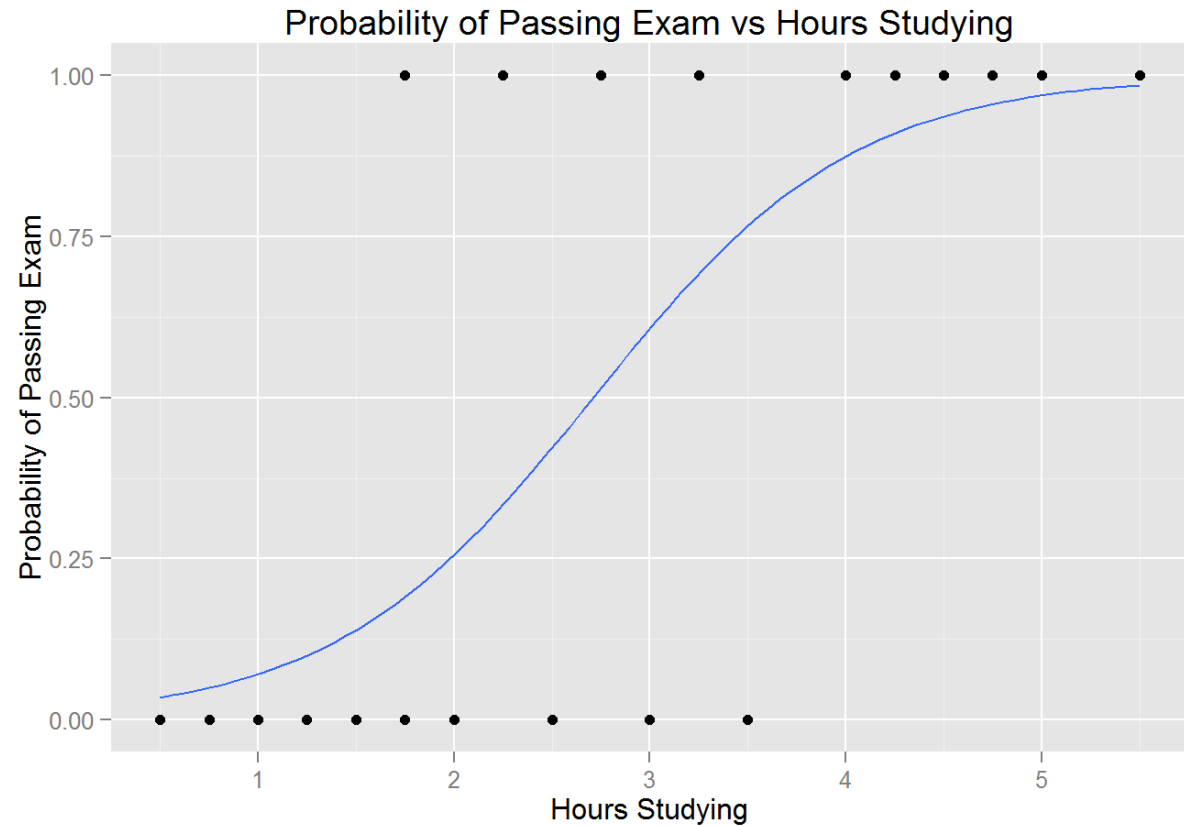
Hours	0.50	0.75	1.00	1.25	1.50	1.75	1.75	2.00	2.25	2.50	2.75	3.00	3.25	3.50	4.00	4.25	4.50	4.75	5.00	5.50
Pass	0	0	0	0	0	0	1	0	1	0	1	0	1	0	1	1	1	1	1	1

Passing exam example

- 20 students
 - Spending between 0 and 6 hours studying for exam
 - Does hours spent studying affect probability of passing?
 - Pass = 1
 - Fail = 0
- 
- 

Logistic regression curve

Graph shows probability of student passing the exam based on amount of hours spent studying



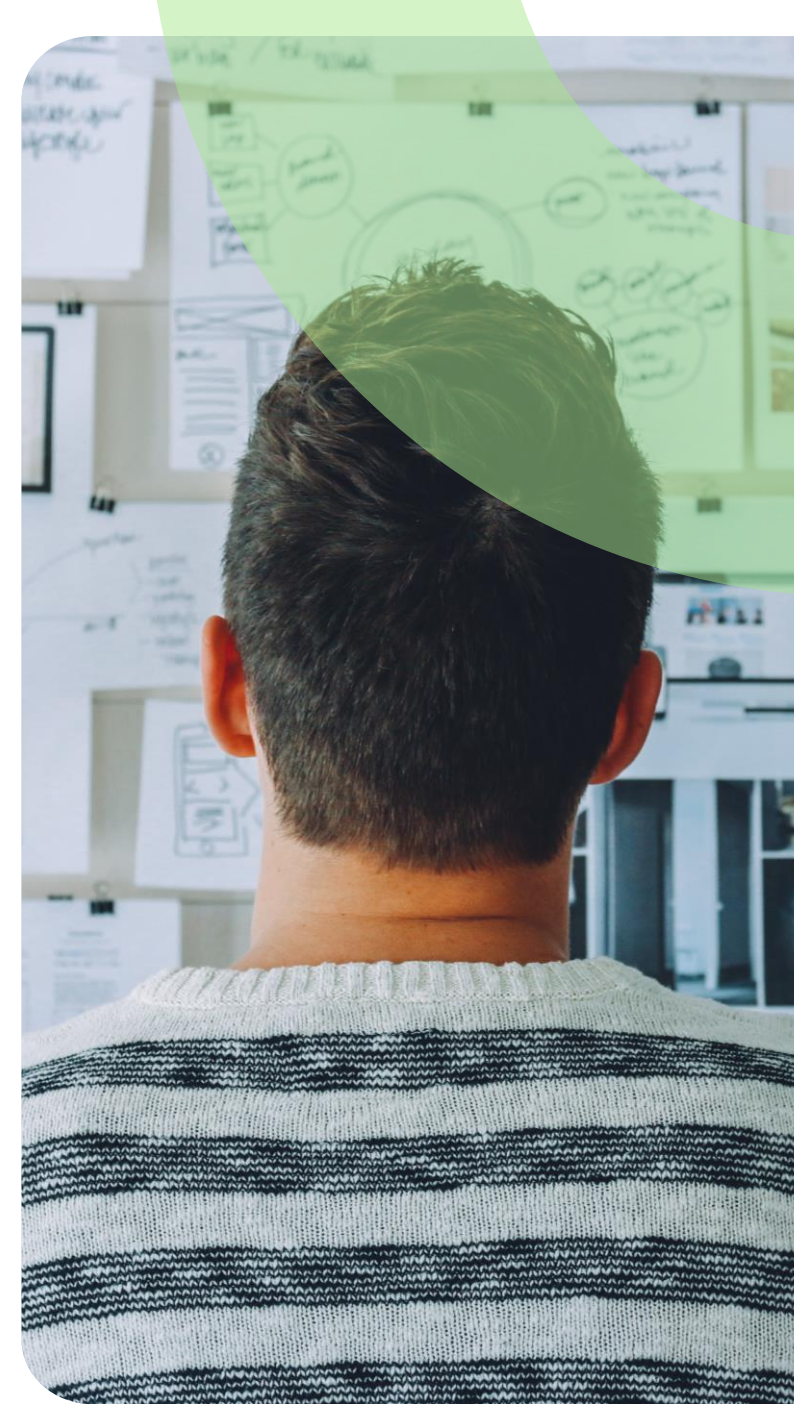
Source: <https://rpubs.com/johnakwei/167443>

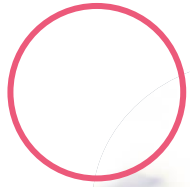
Results

- Hours spent studying is significantly related to probability of student passing exam
- *probability of passing exam =*

$$\frac{1}{1 + e^{-(-4,078 + 1,505(\text{hours studying}))}}$$

- Interpretation: 1 extra hour of study increases log-odds ratio by 1,505





Assumptions

- Binary logistic regression
 - Dependent variable should be binary
- Ordinal logistic regression
 - Ordinal dependent variable
- Independent observations
 - Not repeated or matched data



Assumptions

- Little multicollinearity should exist
- Linearity exists between independent variables and log odds
- Note: needs at least 10 observations for the smallest class





Logistic regression in R

- `glm()`
- `family = binomial`
 - Difference between positive and negative class



Challenge >>

What is the best model for the Boston data set?

#exploredata